

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                 | Marks available |        |     |       |       |      |
|----------|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                 | AO1             | AO2    | AO3 | Total | Maths | Prac |
| 4        | (a) | (i)   | Minimum answer = force per unit charge (accept force per coulomb)                                                                                                                                                                                                                                                                                               | 1               |        |     | 1     |       |      |
|          |     | (ii)  | Use of field equation $E = \frac{kQ}{r^2}$ (1)<br>Distance = $0.6\sqrt{2}$ <b>OR</b> $\frac{1.2}{\sqrt{2}}$ <b>OR</b> 0.85 [m] <b>OR</b> equivalent with sin <b>OR</b> cos45 (1) accept $r^2 = 0.72$ [m <sup>2</sup> ]<br>Penultimate step seen <b>OR</b> 524 5XX seen for field (1)<br>N.B. $k = 9 \times 10^9$ gives exactly 525 000                          | 1               | 1<br>1 |     | 3     | 2     |      |
|          |     | (iii) | Left (independent) <b>or</b> arrow pointing left (1)<br>$\div \sqrt{2}$ seen <b>or</b> $\times \sin 45$ <b>or</b> $\times \cos 45$ (1)<br>Final answer correct = 1 485 000 [N C <sup>-1</sup> ] (1)                                                                                                                                                             |                 | 3      |     | 3     | 2     |      |
|          | (b) | (i)   | Minimum answer – {work done / energy required} per unit charge from infinity<br>(accept {work done / energy required} per coulomb from infinity)                                                                                                                                                                                                                | 1               |        |     | 1     |       |      |
|          |     | (ii)  | [Potential of] +ve and -ve charges cancel (1)<br>Because equidistant <b>OR</b> because the same distance (1) accept P is at the centre<br><b>OR</b> mathematical:<br>Potential equation used (1)<br>An expression for 4 potentials seen to reduce to zero ( $k$ not needed) (1)                                                                                 | 1               | 1      |     | 2     |       |      |
|          |     | (iii) | Right and left force confirmed / agreed with (however stated e.g. accelerates right then decelerates is ok) (1)<br>One of the two forces explained (we'll allow 2 <sup>nd</sup> by implication) e.g. initially repulsed by -ve <b>OR</b> left force later because (closer to) +ve charges (1)<br><br>$E = 0$ at infinity stated <b>or</b> explained somehow (1) |                 |        |     |       |       |      |
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                 |                 |        | 3   | 3     |       |      |

| Question |  |      | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks available |          |          |           |          |          |
|----------|--|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          |  | (iv) | Potential equation used (e.g. 630 000, -282 000) (1)<br>Potentials added and subtracted (696 000) (1)<br>WD = $q\Delta V$ used ( $\pm 0.0264$ ) (1)<br>Energy (not potential) equated to $\frac{1}{2}mv^2$ (1)<br>Final answer ( $8.3 \text{ m s}^{-1}$ ) (1)<br><b>Alternative:</b><br>Potential equation used (-0.02394, +0.0107) (1)<br>PEs added and subtracted (1)<br>Value correct ( $\pm 0.0264$ ) (1)<br>Equated to $\frac{1}{2}mv^2$ (1)<br>Final answer ( $8.3 \text{ m s}^{-1}$ ) (1)<br><br><b>Award a maximum of 3 marks for:</b><br>Any attempt at $a = \frac{F}{m}$ (1) expect $a = 73.3$<br>Any attempt at use of equations of motion (1) i.e. $v^2 = u^2 + 2ax$<br>Answer of $9.4 \text{ [m s}^{-1}]$ (1)<br><br><b>Answers that score 4 marks:</b><br>5.86 $[\text{m s}^{-1}]$ , 2 charges instead of 4<br>Power of 10 – cm instead of m should be 83 $[\text{m s}^{-1}]$ , same applies to mC instead of nC – out by a factor of 1000.<br><br>Unfortunately, nC instead of $\mu\text{C}$ will be out by a factor of $\sqrt{1000}$<br>= 31.6 $[\text{m s}^{-1}]$<br>7.46 $[\text{m s}^{-1}]$ and 11.15 $[\text{m s}^{-1}]$ | 1               |          |          |           |          |          |
|          |  |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1               | 1        |          | 5         | 3        |          |
|          |  |      | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>6</b>        | <b>9</b> | <b>3</b> | <b>18</b> | <b>7</b> | <b>0</b> |